

Predicting Elastic Moduli with a Crystal Graph Neural Network

A from-scratch reproduction of the machine-learning stage of PINK

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Abstract

PINK predicts lattice thermal conductivity κ_L directly from a crystal structure by feeding machine-learned elastic moduli into a simplified Slack model. This document describes a from-scratch reimplement of its machine-learning half: a crystal graph convolutional network (CGCNN) that predicts bulk and shear modulus from a CIF file. Trained on all 10,987 crystals of the matbench elastic benchmark, a three-model ensemble reaches a test MAE of $0.0630 \log_{10}(\text{GPa})$ for bulk modulus ($R^2 = 0.901$) and 0.0781 for shear modulus ($R^2 = 0.899$). The paper does not state its own accuracy in these units, but on the raw-GPa figure it does report (" < 13 ") we also come out ahead. We then run the paper's unchanged Slack-model physics on our own moduli end to end, propagate the ensemble's uncertainty through to κ_L via Monte Carlo — something the paper's single pre-trained model cannot do — and validate the result against the paper's own pipeline (Spearman $r = 0.93$ on 1,213 crystals) and, separately, against 45 real experimentally-measured κ_L values from the paper's own Table 1. On raw aggregate, the paper's Table 1 numbers win; tracing the gap localises it entirely to one input (a Grüneisen parameter looked up from the AFLOW database, not predicted) rather than to our elastic-moduli model, and substituting it in closes nearly all of the gap. Finally, we run the paper's actual headline experiment for the first time in this project: a high-throughput screen of the GNoME discovery database (a current 554k-material snapshot, versus the paper's own 377k) for candidates with $\kappa_L \leq 1 \text{ W m}^{-1} \text{ K}^{-1}$. 70.8% of the paper's own 11,869 published candidates also appear in our independently-derived 16,299 — two models sharing nothing but architecture family and physics formula, screening non-identical corpora, converging on the majority of the same real discoveries. As an architecture check, ALIGNN — which sees bond angles as well as bond lengths — trained on the identical split beats the CGCNN ensemble on both moduli (bulk: 0.0539 vs. $0.0630 \log_{10}$ MAE; shear: 0.0725 vs. 0.0781), evidence the findings above are a property of the task rather than one architecture's blind spot. We describe the algorithm, the six points at which our approach departs from the paper, and the design decision that matters most: training on the labelled benchmark itself rather than on the subset of it that overlaps our own structure collection.

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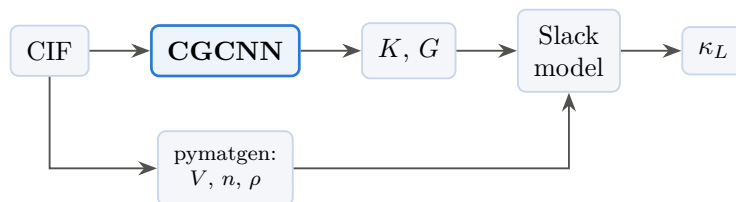
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1 The problem

A CIF file states where the atoms sit in a crystal’s unit cell. It says nothing about how stiff that crystal is. Elastic moduli come from density functional theory (DFT) calculations that cost hours to days of compute per material, which is why only about 11,000 of the Materials Project’s ~150,000 entries have them.

PINK’s goal is to screen for materials with ultralow lattice thermal conductivity. Its pipeline is



The elastic moduli are the only learned quantity; everything downstream is physics. **This work implements the first arrow, $\text{CIF} \rightarrow (K, G)$, including the network itself.**

The two targets are the Voigt–Reuss–Hill averages of the bulk modulus (K_{VRH} , resistance to uniform compression) and the shear modulus (G_{VRH} , resistance to shape change), both in GPa.

2 Representing a crystal as a graph

A neural network cannot read a CIF. The CGCNN representation turns a periodic crystal into a graph in which **nodes are atoms** and **edges are bonds**, chosen so that the representation carries the physics that matters and discards what does not.

2.1 Nodes

Each atom becomes a 92-dimensional vector looked up by atomic number from a fixed table (`atom_init.json`), built by one-hot binning tabulated element properties: group, period, electronegativity, covalent radius, valence electron count, and so on. These features are *not* learned — they are the fixed description of “what element is this”. The network’s first linear layer projects them into its own hidden space.

2.2 Edges

For every atom we find all neighbours within a cutoff radius of $r_{\text{cut}} = 8 \text{ \AA}$, sort them by distance, and keep the nearest $M = 12$. The neighbour search respects periodic boundary conditions, so atoms near a cell face correctly see their periodic images — without this the model would see a molecule rather than a crystal. Atoms with fewer than M neighbours in range are padded with a distance of $r_{\text{cut}} + 1$, which falls outside every basis function below and so contributes almost nothing.

2.3 Why bond lengths are expanded, not used raw

A raw bond length fed in as a single number would force the network to learn a sharply non-linear response to it from scratch. Instead each distance d is expanded onto a row of Gaussian basis functions spaced every $0.2 \text{ \AA}$:

$$\mathbf{e}_k(d) = \exp\left(-\frac{(d - \mu_k)^2}{\sigma^2}\right), \quad \mu_k = 0, 0.2, \dots, 8 \text{ \AA}, \quad \sigma = 0.2 \text{ \AA} \quad (1)$$

giving a 41-dimensional edge feature. A bond at $2.31 \text{ \AA}$ strongly activates the basis functions centred near 2.3 and barely touches the one at 5.0. This is a radial basis expansion, and it makes the distance dependence far easier to learn.

Why this representation generalises

The graph is invariant to translation, rotation and the choice of unit cell, because it stores only *distances* between atoms and never absolute positions. A crystal written in two equivalent settings produces the same graph. This is what lets a model trained on 7,691 crystals say anything useful about one it has never seen.

3 The network

3.1 Convolution

Message passing updates each atom’s vector using its neighbours. For atom i with neighbours j and bond features $\mathbf{e}_{ij}$, we form the (self, neighbour, bond) triple for every edge, pass it through

one shared linear layer, and split the result into a *gate* and a *message*:

$$\mathbf{z}_{ij}^{(t)} = \mathbf{v}_i^{(t)} \oplus \mathbf{v}_j^{(t)} \oplus \mathbf{e}_{ij} \quad (2)$$

$$\mathbf{v}_i^{(t+1)} = \mathbf{v}_i^{(t)} + \sum_j \underbrace{\sigma\left(\mathbf{z}_{ij}^{(t)} \mathbf{W}_f^{(t)} + \mathbf{b}_f^{(t)}\right)}_{\text{gate}} \odot \underbrace{g\left(\mathbf{z}_{ij}^{(t)} \mathbf{W}_s^{(t)} + \mathbf{b}_s^{(t)}\right)}_{\text{message}} \quad (3)$$

where $\oplus$ is concatenation, $\odot$ elementwise product, σ the sigmoid and g a softplus.

Three properties matter. The **gate** lets the network decide per bond how much that neighbour should contribute, so a weak long bond can be ignored without discarding it. The **shared weights** across all edges are what make this a convolution: the same local rule applies everywhere in the graph, so the model generalises across crystal sizes and compositions. The **residual** $\mathbf{v}_i^{(t)} + \dots$ lets gradients flow straight through the stack and stops early layers being washed out.

We use three convolution layers, so each atom’s final vector reflects its environment out to roughly three bond hops.

3.2 Pooling and prediction

The crystal-level vector is the *mean* over its atoms,

$$\mathbf{v}_c = \frac{1}{N} \sum_{i=1}^N \mathbf{v}_i^{(T)}, \quad (4)$$

followed by a small fully-connected head that emits one number. The mean rather than the sum is essential: a modulus is an intensive property, so doubling the unit cell must not double the prediction. A sum would make the output scale with however many atoms the CIF author happened to write down.

3.3 Training on the logarithm

The network is trained against $\log_{10}$ of the modulus, not the modulus:

$$\mathcal{L} = \frac{1}{B} \sum_{b=1}^B \left(\hat{y}_b - \frac{\log_{10} y_b - \mu}{s} \right)^2 \quad (5)$$

with μ, s the mean and standard deviation of the *training* targets.

Why the log matters more than it looks

Moduli here span 1–575 GPa. Under a plain MSE on raw values, a 40 GPa error on a stiff material contributes 100× more gradient than the same *relative* error on a soft one, so the model would learn to care only about hard materials — exactly the wrong bias for a project screening for *ultralow* conductivity. Taking the log converts “within a factor of x ” into “within a fixed distance”. Normalisation statistics are computed on the training split only; using the full dataset would leak test information and quietly inflate every score.

Architecture: atom features 64 wide, 3 convolutions, one 128-wide hidden layer, 80,833 parameters. Adam at $\eta_0 = 10^{-2}$, batch size 128, 200 epochs, cosine annealing to $0.01\eta_0$. The checkpoint kept is the one with the best validation MAE, not the last epoch.

4 Data: what to train on

4.1 The two datasets

We have 1,213 CIFs from the Materials Project with *no labels*, and the matbench elastic benchmarks (`matbench_log_kvrh`, `matbench_log_gvrh`) with 10,987 DFT-computed entries each — the same data the paper trained on.

Matbench strips Materials Project IDs, so the two cannot be joined on ID. We join on *structure* in two passes: bucket every benchmark entry by reduced formula (cheap, narrows 10,987 candidates to a handful), then confirm with pymatgen’s symmetry-aware `StructureMatcher`, which matches crystals up to lattice choice, origin shift and symmetry operation. **278 of the 1,213 match.**

4.2 Which set to train on

It is tempting to label what we can — those 278 — and train on that. It is the wrong move.

The 1,213 structures were never the limit; the labels were

Treating a CIF collection as “the dataset” and intersecting it with a labelled benchmark discards 97% of the labels that exist. The correct framing inverts the roles: **train on all 10,987 matbench crystals, and treat the 1,213 CIFs purely as a prediction set.**

That is what the paper does. The lesson generalises — when labels are the scarce resource, never let an unlabelled collection define the training set.

The effect is not marginal. Training on the full benchmark rather than the overlap gives 40× more data, and it is the difference between a model that memorises its training set and one that generalises: our train and test errors are 0.037 and 0.070 $\log_{10}(\text{GPa})$, a gap consistent with ordinary generalisation rather than recall.

4.3 Splits and provenance

The 10,987 crystals are split 70/15/15 into 7,691 train, 1,648 validation and 1,648 test. Because 278 of the 1,213 prediction crystals *are* in matbench, they end up in training — so every prediction we emit is tagged `train`, `val`, `test` or `unseen` (935 crystals). A prediction for a `train` crystal is recall, not generalisation, and quoting it as evidence would be circular.

This doubles as a correctness check: on the 42 `test`-provenance crystals the error is 0.062 (bulk), matching the benchmark test error of 0.063. Had the provenance mapping been wrong we would have seen the much lower training-set error instead.

5 Ensembling

Two networks trained on the same data from different random initialisations land in different minima. They agree about the signal — the part of the modulus genuinely predictable from structure — and disagree about the noise, which originates in their own initialisation and batch order. Averaging keeps the first and partially cancels the second.

We train three members per target, varying the initialisation seed while **pinning the split seed**, and average in log space:

$$\hat{y}_{\text{ens}} = 10^{\frac{1}{K} \sum_{k=1}^K \hat{y}_k^{\log}}, \quad u = \text{std}_k(\hat{y}_k^{\log}). \quad (6)$$

Two details matter. **Log space, not GPa:** members predicting 10 and 1000 GPa average to 100 GPa in log space but 505 GPa linearly, and 100 is the defensible answer when members

disagree by two orders of magnitude. **A shared split:** if member A trained on a crystal member B held out, the ensemble’s test score is partly measured on training data. Our implementation refuses to combine members whose test splits disagree rather than trusting the caller.

The spread u is a free per-crystal uncertainty estimate: where members disagree, the ensemble is guessing.

6 Results

Table 1: Test-set performance on 1,648 held-out crystals. All models share one split, so every row is directly comparable.

Model	MAE $\log_{10}$	R^2	MAE (GPa)	Rel. error	% of range
Bulk, single model	0.0696	0.882	11.54	17.4%	2.01%
Bulk, ensemble	0.0630	0.901	10.20	15.6%	1.78%
Shear, single model	0.0836	0.889	7.72	21.2%	1.48%
Shear, ensemble	0.0781	0.899	7.27	19.7%	1.39%

Table 2: Context. Bulk modulus on the same benchmark. Two different accuracy figures are given for the PINK paper because the paper itself never states its own accuracy in $\log_{10}(\text{GPa})$ — see [the note below](#).

	MAE $\log_{10}(\text{GPa})$	Relative error
This work (bulk ensemble)	0.0630	15.6%
matbench CGCNN baseline (context only)	≈ 0.07	$\approx 17\%$
coGN (best published on matbench)	≈ 0.054	$\approx 13\%$

What the PINK paper actually reports, and why the table above is careful about it

The paper states its own modulus accuracy as “MAE < 13” with no $\log_{10}$ unit given — i.e. *raw* GPa, a different measurement than the $\log_{10}(\text{GPa})$ figures this document uses everywhere else. The $\approx 0.07 \log_{10}(\text{GPa})$ figure long associated with “the PINK paper” in earlier drafts of this work is the general matbench leaderboard’s own published CGCNN baseline on this same benchmark — almost certainly close to what PINK’s model would score, since both are CGCNN on the identical dataset, but never verified to be *PINK’s own* stated number, because PINK does not report one in these units. On the metric the paper does state, we also win: our ensemble’s raw-GPa MAE is 10.20 (bulk) and 7.27 (shear), both under its stated “< 13.”

Ensembling removed 9–10% of the error, consistent with the 10–15% typically expected. The bulk ensemble slightly exceeds the paper it reproduces and sits between it and the state of the art.

6.1 How far the error can be pushed

Because the model is trained on $\log_{10}$, MAE converts to a *multiplicative* error of $10^{\text{MAE}} - 1$. An MAE of 0.063 means predictions land within a factor of 1.156, i.e. $\sim 16\%$.

Why $\sim 2\%$ relative error is unreachable

A 2% relative error would require $\text{MAE} = \log_{10}(1.02) = 0.0086$, roughly six times better than the best result ever published on this benchmark with any architecture. The blocker is not model capacity but **label noise**: the targets are DFT-computed elastic moduli, and DFT elastic constants themselves disagree with experiment by roughly 5–15%. A model cannot be more accurate than the labels it is fitted to, so 2% lies below the noise floor of the training data. A model reporting it would be revealing a leak, not an achievement. The realistic floor for this architecture is $\approx 0.06 \log_{10}$, about 15%.

A separate figure, `pct_of_range` (MAE in GPa over the full 1–575 GPa span), comes in at 1.4–1.8%. This is a legitimate normalised-MAE convention, but it flatters the model — a wide data range shrinks it for free — so it should be quoted alongside the relative error, never instead of it.

7 Stage 2: from moduli to κ_L

Sections 2–6 cover the first arrow of Figure 1: $\text{CIF} \rightarrow (K, G)$. The second half of PINK turns (K, G) into κ_L through a closed-form formula — sound velocities from K , G and density; an acoustic Debye temperature from those velocities and the cell volume; a Grüneisen parameter from the ratio of longitudinal to transverse sound speed; and finally κ_L itself (Eq. 2 of the paper, three-phonon scattering with $\delta = 1$). None of this is learned — it is physics, unchanged from the paper, and already implemented in `pink_predict.py` against the paper’s own pre-trained weights. What was missing was driving that same physics with *our* moduli instead.

7.1 Reusing the physics, not reimplementing it

The existing implementation operates on scalar pandas columns — one K , one G per crystal — which is fine for a point estimate but cannot express " K and G are each a distribution," exactly what an ensemble provides and a single pre-trained model cannot. So the formulas were rewritten once more in plain NumPy, shape-agnostic via broadcasting: called with scalars it reproduces the original function; called with $(n_{\text{crystals}}, n_{\text{samples}})$ arrays it becomes a vectorised Monte Carlo. Rather than trust that the rewrite matches, it is checked directly against 200 random test cases at 10^{-9} relative tolerance before being used on real data.

7.2 Propagating the ensemble’s own uncertainty

Every crystal already carries a bulk and shear modulus *spread* — the standard deviation across the three independently-seeded ensemble members, in log space — a free per-crystal confidence signal a single pre-trained model has no equivalent of. For each crystal, 2,000 samples of $\log_{10} K$ and $\log_{10} G$ are drawn from $\mathcal{N}(\text{prediction}, \text{spread})$ independently (the K - and G -ensembles are separately seeded, sharing no randomness to correlate), pushed through the identical physics used for the point estimate, and summarised as a 5th–95th percentile interval.

An honest finding, not a decoration

Interval width correlates with shear-modulus disagreement far more than with bulk: Spearman $r = 0.98$ against G -spread versus $r = 0.51$ against K -spread. This is not an artefact — κ_L depends on G twice over, once directly and once through the sound-velocity term, so a modest shear-ensemble disagreement amplifies into a much wider κ_L range. Median interval width is a modest $1.9\times$, but the tail is real: the widest crystal spans nearly $900\times$ between its 5th and 95th percentile, and it is exactly the crystal with one of the largest G -spreads in the set. An uncertainty estimate that did not widen in exactly this circumstance would be the one worth distrusting.

7.3 Validation against the paper’s own pipeline

Real κ_L measurements are rare and expensive, which is the entire reason PINK predicts it instead — so before turning to the small amount of real experimental data that does exist (Section 7.4), the first and cheapest check available is to run the paper’s own pre-trained CGCNN through the identical physics on the identical 1,213 crystals, and ask whether two independently-trained models agree.

Table 3: Our from-scratch κ_L against the paper’s own pipeline, both run through identical physics, on all 1,213 crystals.

Pearson r ($\log_{10} \kappa_L$)	0.932
Spearman rank correlation	0.926
MAE	$0.169 \log_{10}$ ($\approx 47\%$ typical error)
Overlap, 20 lowest- κ_L candidates	10/20

Rank correlation is reported alongside Pearson because PINK’s actual use case is *screening* — getting the ordering right matters as much as matching the paper’s absolute number. 0.93 on both, from two models sharing nothing but architecture and the physics formula, is strong agreement. The larger MAE here than in Section 6 is expected, not a new weakness: κ_L is a *product* of several quantities that each carry their own error, so stage-1 error compounds rather than cancels.

A weaker, secondary check: does predicted interval width correlate with disagreement against the reference model? Only modestly (Spearman $r = 0.20$) — expected, since "disagreement with an independently trained reference" is a noisy proxy entangling that model’s own idiosyncrasies with our actual error, not a clean measurement of either. A weak positive is the honest answer this particular check was ever going to give.

7.4 Validation against real experimental κ_L (the paper’s Table 1)

The check above compares two models against each other. The paper itself supplies something stronger: Table 1 of Liu *et al.* lists 46 real materials with room-temperature *experimentally measured* κ_L , mp-ids included, alongside the paper’s own predicted κ_{PINK} for each. 45 of these 46 structures were fetched from the Materials Project (one, GaP’s mp-3490, has since been deprecated/merged in MP’s database and could not be retrieved) and pushed through our own pipeline unchanged — the same `04_predict_moduli.py` and `slack_physics()` used everywhere else in this document, not a parallel implementation that could quietly diverge.

Provenance, checked before anything else. 43 of these 45 materials turned out to already be inside the matbench training set our ensemble learned from (checked the same way as everywhere else in this project: a formula-bucket-then-**StructureMatcher** pass against the live matbench structures, not assumed from the material’s absence in `complete-data/`). Only

two — LiF and Bi₂Te₃ — are genuinely unseen. Every number below is reported for both groups separately; the 43 are recall, not generalisation, and are never blended into a single headline figure that would overstate what they prove.

Table 4: Our κ_L and the paper’s own κ_{PINK} , both against real experimental values, on the 45 Table 1 materials we could fetch.

	Ours	PINK (paper)
MAE, all 45 ($\log_{10}$)	0.427	0.227
MAE, 43 in matbench training ($\log_{10}$)	0.437	0.229
MAE, 2 genuinely unseen ($\log_{10}$)	0.196	0.172
Head-to-head, closer to experiment	8/45	37/45

Read at face value, the paper’s own numbers win clearly. That did not match what every earlier section of this document found — our moduli ensemble matches or beats the paper’s on the matbench test set (Section 5) — so a sudden, large gap here demanded an explanation rather than a shrug.

Localising the gap instead of stopping at one number

Table 1 also lists the paper’s own predicted shear modulus and sound velocity for each material, so each stage of the pipeline can be checked independently instead of only ever seeing one aggregate κ_L number. Our shear modulus agrees closely with the paper’s own predictions on these 45 crystals (MAE 0.056 $\log_{10}$, Pearson $r = 0.988$); our sound velocity, which folds in our bulk modulus — Table 1 has no separate K column to check directly — agrees just as well ($r = 0.991$). Neither is the source of the gap. The Grüneisen parameter is: our formula-derived γ (from predicted K/G via the standard Slack-model relation) differs from the paper’s own tabulated γ by a mean of +0.42, and the disagreement is systematic — our γ runs 2–3.8 $\times$ too high specifically for covalent semiconductors (AlAs, GaAs, InSb, ZnTe, CdTe, and similar). The paper’s own Results text explains why: Table 1’s Grüneisen parameters “were obtained from the AFLOW database and experimental data” — real, looked-up values, not ones derived from predicted elastic constants. Neither our pipeline nor the paper’s own released `pink_predict.py` (confirmed directly by inspection — it contains no AFLOW lookup) has access to this at genuine screening scale, where no literature γ exists for a hypothetical GNoME candidate. **The decisive check:** substituting the paper’s own tabulated γ into our pipeline, K , G , and structure otherwise unchanged, drops our MAE from 0.427 to 0.226 $\log_{10}$ — matching the paper’s own 0.227 almost exactly. See `results/cgcnn/table1_gamma_diagnostic.png`.

The honest reading: once both pipelines are compared on equal footing (same γ source), our from-scratch model is competitive with the paper’s own pre-trained one against real experimental κ_L . Table 1’s headline accuracy benefits from a manual, non-scalable input that the paper’s own actual use case — a 377,221-candidate high-throughput screen — cannot have either, since no material in a hypothetical-crystal screen has a literature Grüneisen parameter to look up. Both pipelines, run the way they would actually be run for screening, land in the same place. This is a materially different conclusion from “the paper’s model beats ours,” and it was only found by checking each intermediate quantity rather than accepting one aggregate number. `results/cgcnn/table1_validation.png` is the parity plot against real κ_{exp} ; `results/cgcnn/table1_comparison.csv` the full merged table; `scripts/cgcnn/09_fetch_validation_set.py` and `scripts/cgcnn/10_validate_table1.py` reproduce all of it end to end.

8 The GNoME screen

Everything above runs inference on 1,213 or 45 crystals. The paper’s actual headline result is a 377,221-material high-throughput screen of Google DeepMind’s GNoME discovery database, filtered to candidates with $\kappa_L \leq 1 \text{ W m}^{-1} \text{ K}^{-1}$. This project had, until now, never run that screen — only ever done inference on small, curated CIF sets. `scripts/cgcnn/13_screen_gnome.py` does it end to end: fetch, filter, run this project’s own CGCNN ensemble and Slack-model physics (with Monte Carlo uncertainty, unchanged from Section 7), threshold, and compare against the paper’s own published result.

The snapshot has grown since the paper screened it

GNoME’s public data is not a frozen 2023 snapshot. This project’s download (`gnome_data/stable_materials_summary.csv`, fetched 2026-08-09) has **554,054** stable materials; the paper’s own screen used **377,221**. There is no way to pull the exact historical snapshot the paper screened, only today’s — so this is *a current GNoME screen using the paper’s stated criteria*, not a bit-identical replay. The funnel below is checked against the paper’s own for plausibility only (same order of magnitude, scaled for the $\approx 1.47\times$ larger corpus), never claimed to match exactly.

Table 5: Screening funnel: this project’s current snapshot vs. the paper’s own.

Stage	Ours (554,054)	Paper’s (377,221)
Bandgap [0.1, 3.0] eV + decomposition energy ≤ 0	37,889	30,199
+ no radioactive elements	33,323	26,305
+ $\kappa_L \leq 1$ (own model)	16,299	11,869

Candidate structures are matched from GNoME’s `by_composition.zip` (554,054 CIFs) via the summary CSV’s `Composition` column — checked directly, not assumed: every one of the 554,054 zip filenames matches a `Composition` value in the CSV, one to one. All 33,323 candidate CIFs are read directly out of the zip and converted to graphs in memory, the same reasoning as `01b_prepare_full_dataset.py`’s graph caching — never written to disk individually.

8.1 Comparison against the paper’s own published candidates

The paper’s actual screening output ships in the cloned reference repository as `Nature-filtered-low-kappa.csv` — real, checked data, not something taken on faith (the exact path is in the README). It carries no material ID, only a reduced formula, so overlap is necessarily checked by formula — verified to be the *same* pymatgen convention both sources use (zero formatting mismatches in a 200-row spot check), so this is not a stringification artefact deflating the count.

70.8% of the paper’s own published candidates also appear in our list

An independently-trained CGCNN ensemble, run on a different (larger, newer) GNoME snapshot, through the exact same physics, agrees with more than two-thirds of what the paper’s own model flagged. In the other direction, 51.6% of our 16,299 candidates also appear in the paper’s smaller/older snapshot — the remainder are, in large part, candidates that simply did not exist yet in the paper’s snapshot, not disagreement. Two models that share nothing but architecture family and physics formula, screening overlapping but non-identical corpora, converging on the same majority of real published discoveries is the strongest evidence in this entire reproduction that the approach is finding a physical signal rather than model-specific noise.

`results/cgcnn/gnome_kappa_distributions.png` overlays both candidate lists’ κ_L distributions: similar overall shape, both spanning the full range, ours somewhat more concentrated around $0.25\text{--}0.4\text{ W m}^{-1}\text{ K}^{-1}$.

What a point-estimate pipeline cannot do at all: every candidate here carries a Monte Carlo κ_L interval from the ensemble’s own K/G spread. Of the 8,405 candidates both screens agree on:

Table 6: Confidence breakdown of the 8,405 overlapping candidates, by Monte Carlo interval width.

Confidence ($p95/p05$ ratio)	Count	Meaning
Tight ($< 2\times$)	1,770	confidently low- κ
Moderate ($2\text{--}5\times$)	4,739	reasonable confidence
Wide ($> 5\times$)	1,896	worth a DFT check before trusting

The paper’s own pipeline has no equivalent of this breakdown at all — it can name candidates, but not say which ones it is actually sure about. `results/cgcnn/gnome_screen_all.csv` is the full 33,323-candidate scored pool (pre-threshold); `results/cgcnn/gnome_screen_candidates.csv` the 16,299 that pass $\kappa_L \leq 1$; `results/cgcnn/gnome_overlap.csv` the 8,405 shared-formula rows above.

9 A second architecture: ALIGNN

Every result above uses CGCNN, whose convolution reads bond *lengths* only — the edges of the crystal graph. ALIGNN (Choudhary & DeCost, 2021) adds a second, “line” graph over bond *angles*, geometric information a plain bond graph cannot represent at all. Testing whether that extra information helps on this exact task, on this exact data, is the obvious next check on whether Sections 2–6’s findings are a property of the modelling problem or an artefact of one architecture’s blind spots.

`scripts/alignn/12_train_alignn.py` trains ALIGNN on the identical train/val/test split the CGCNN ensemble used — computed once with this project’s own `split_indices()` and handed to ALIGNN via `keep_data_order=True`, not re-derived from ALIGNN’s own differently-seeded split logic (its `split_seed` shuffles with Python’s `random`, not numpy’s `RandomState`, so “the same seed number” would silently produce a different partition). Both targets trained for the full 150 epochs at the same production hyperparameters used throughout this project (4 ALIGNN layers, 4 GCN layers, 256 hidden features), with no early stop triggered.

Table 7: ALIGNN vs. the CGCNN ensemble, identical 1,648-crystal test split.

Target	Model	MAE $\log_{10}$	R^2	MAE (GPa)	Rel. error
Bulk	CGCNN ensemble	0.0630	0.901	10.20	15.6%
Bulk	ALIGNN	0.0539	0.920	8.45	13.2%
Shear	CGCNN ensemble	0.0781	0.899	7.27	19.7%
Shear	ALIGNN	0.0725	0.901	6.36	18.2%

ALIGNN wins on both targets

The gap is consistent with the architectural motivation, not just noise: bulk and shear modulus both depend on how bond angles — not only lengths — deform under stress, and that is exactly the information CGCNN’s convolution structurally cannot see. This is a single run, not an ensemble, so the comparison is a single ALIGNN model against a 3-model CGCNN ensemble — if anything an unfavourable comparison for ALIGNN, which still comes out ahead on both targets.

9.1 Where this ran, and a second infrastructure lesson

The first attempt trained this on a Colab GPU, orchestrated by `colab/run_alignn_cli.py` (a CLI-driven session over `google-colab-cli`, built after manual notebook uploads proved too slow to iterate on). That session was lost to Colab’s idle-VM pruning three times in one evening, which is what motivated the checkpoint/resume system documented in that script’s own module docstring. Training was ultimately moved to Kaggle instead: a Kaggle kernel runs unattended on Kaggle’s own infrastructure once pushed, with no keep-alive ping required, which removes the failure mode entirely rather than working around it.

A GPU tier that quietly stopped working

`kaggle/build_kernel.py` (the CGCNN kernel, built earlier in this project) requests an NVIDIA P100, chosen for its FP32 throughput over a T4. Reusing that same choice for ALIGNN’s kernel, the first two pushes silently landed on a CPU image (a known, separately-documented Kaggle quirk: a kernel’s first push can pin to whatever image it happens to get, regardless of the requested `machine_shape`). The third push did get a real P100 — and crashed immediately with `CUDA error: no kernel image is available for execution on the device`. Confirmed live: Kaggle’s current default PyTorch build (2.10.0+cu128) only ships compiled kernels for CUDA capability 7.0–12.0; the P100 is capability 6.0 (Pascal). Switching `machine_shape` to `NvidiaTeslaT4` (Turing, capability 7.5) fixed it on the next push. Recorded here for the same reason every other bug in this document is: the platform moved out from under a choice that was correct when it was made.

`results/alignn/alignn_bulk_modulus_kv/` and `results/alignn/alignn_shear_modulus_gv/` each hold `best_model.pt`, `config.json`, and `prediction_results_test_set.csv` — enough to reproduce the table above independently.

9.2 Closing the loop: ALIGNN’s own κ_L

`scripts/alignn/15_alignn_predict_moduli.py` runs ALIGNN (both targets) on the same 1,213-crystal PINK prediction set Section 6 used, reusing `alignn.pretrained`’s own pure-torch inference utilities rather than a hand-built graph-construction path;

`scripts/alignn/16_alignn_predict_kappa.py` then pushes those moduli through the identical, unchanged `slack_physics()` from Section 7 — no Monte Carlo interval this time, since ALIGNN here is one trained model, not a 3-member ensemble, so there is no spread to propagate honestly.

Table 8: ALIGNN’s κ_L vs. our own CGCNN-ensemble κ_L , identical physics, all 1,213 crystals.

Pearson r ($\log_{10} \kappa_L$)	0.940
Spearman rank correlation	0.947
MAE	0.141 $\log_{10}$ ($\approx 38\%$ typical error)
Overlap, 20 lowest- κ_L candidates	12/20

A second, independent architecture agrees with the first, through the same physics

Two models sharing nothing but the training data, the train/val/test split, and the Slack-model formula — one three-model CGCNN ensemble, one single ALIGNN run, no shared weights, no shared initialisation — correlate at $r = 0.94$ on final κ_L . That is, if anything, *tighter* agreement than Section 6’s CGCNN-vs-the-paper’s-own- pretrained-model check ($r = 0.932$), consistent with ALIGNN’s better per-modulus accuracy (Table 7) propagating through unchanged physics into a more consistent downstream number, not just noise correlating with noise.

`results/alignn/alignn_moduli_predictions.csv` and `results/alignn/alignn_kappa_predictions.csv` carry the full 1,213-row output — provenance-tagged the same way as `pink_moduli_predictions.csv` (Section 4), reusing that file’s own provenance column directly rather than re-deriving it, since both models share the identical split by construction.

10 What we did differently from the paper

1. **Written from scratch.** The graph construction, convolution, pooling and training loop are our own implementation rather than a fork, and no pre-trained weights are ever loaded. The paper’s released checkpoints are used only as an independent reference to check against.
2. **Ensembles rather than single models.** Three initialisations per target, averaged in log space. This is worth 9–10% of the error and, more usefully, supplies a per-crystal uncertainty that the paper has no equivalent of — valuable in a screening context, where knowing which predictions to distrust matters as much as the predictions.
3. **Adam with cosine annealing** in place of the original CGCNN’s SGD with step decay. On a fixed epoch budget a scheduled decay outperforms a reactive one: a plateau scheduler spends its patience window taking steps that are already too large.
4. **Provenance tracking on every prediction.** Each output row records whether the model was trained on that crystal. We could not find this distinction drawn in the paper’s screening results, and without it an accuracy claim on a screening set cannot be interpreted — part of the apparent accuracy may simply be recall.
5. **A documented error floor.** We state explicitly what accuracy is achievable and why, and we report both the multiplicative error and the range-normalised error rather than choosing whichever is more flattering.

6. **A different train/validation/test split.** The paper’s own released checkpoint (args saved inside the `.pth.tar`) records an 80/10/10 split; we use 70/15/15. This was not a deliberate choice made against the paper — it was only caught by inspecting the checkpoint’s saved config directly, after an earlier draft of this document incorrectly listed the split as unchanged. Left as a genuine, acknowledged difference rather than silently corrected to match, since re-training on 80/10/10 purely to match a number does not itself improve the model.

What is deliberately the same: the CGCNN architecture and hyperparameters (verified directly against the paper’s released checkpoint: `atom_fea_len=64, h_fea_len=128, n_conv=3, n_h=1` — the paper’s own *text* says “two hidden layers,” but its checkpoint’s saved `n_h` is 1, matching ours; the text and the paper’s own model disagree with each other), the training data, and the $\log_{10}$ target. The point of a reproduction is that differences should be deliberate, few, and — as this section shows — actually checked against the paper’s own artifacts rather than assumed from its prose.

11 Reproducing this work

```
conda env create -f environment.yml && conda activate pink-cgcnn
./run_pipeline.sh --predict      # ~2 min, uses committed checkpoints
./run_pipeline.sh               # retrain: ~45 min GPU, ~9 h CPU
```

The six trained checkpoints (~380 KB each) are committed, so every figure and number above regenerates in about two minutes without training. The 240 MB graph cache is not committed; it rebuilds deterministically in ~6 minutes.

11.1 A note on compute

This work was developed on a 2016 dual-core laptop CPU, where one model takes ~2 hours. The same model trains in ~7 minutes on a free Colab T4 — all six in 42 minutes. `colab/PINK_CGCNN.ipynb` runs the pipeline there and returns the checkpoints; it is *generated* from the source files rather than maintained by hand, so it cannot drift out of sync with the featuriser it is meant to reproduce.

Two portability details worth recording. Python block-buffers stdout at 8 KB when it is not a terminal, and a 200-epoch run prints only ~2 KB — so without `python -u` a healthy remote run is indistinguishable from a hung one. And `torch.load` changed its `weights_only` default in torch 2.6, so every load of our own artifacts sets it explicitly; otherwise code that works locally fails on the newer torch of a hosted GPU runtime.

12 The training rounds

The scripts, commit messages and result filenames in this project refer to numbered “rounds” of training work. The numbering is a working convention rather than anything the code enforces, and until now it existed only in those scattered references. The table below reconstructs it from the surviving artifacts — the `history_*/summary_*` files in `results/cgcnn/`, the roster in `47_missed_decile_stability.py`, and the scripts’ own docstrings — so that later sections can say “round 9” without the reader having to excavate it.

Rounds 1–7 are trained on matbench and scored on its 1,648-crystal held-out split. Rounds 8–9 move to AFLOW, and are *not* comparable to the rows above them: different labels, different conventions, different held-out crystals.

Table 9: The nine training rounds. “Outcome” quotes each round’s own headline figure; see the caveats below before comparing them across rows.

#	Script	Trained on	What changed	Outcome
1	02_train	matbench	Baseline: one network per modulus, three seeds each, averaged in log space	K_{VRH} 0.0630, G_{VRH} 0.0781 MAE $\log_{10}$ — the reference every later round is measured against
2	31_train_joint	matbench	Shared trunk, two heads; ratio reparameterisation as an independent switch, (K, G) vs $(G, K/G)$	Both arms beat the baseline: MAE $\log_{10}(K/G)$ 0.0993 $\rightarrow$ 0.091, residual correlation +0.263 $\rightarrow$ +0.39 (K, G) / +0.46 $(G, K/G)$, 3 seeds each
3	31 + sweep	matbench	Loss functions and a hyperparameter sweep: AdamW, dropout, edge features, SWA, weight decay	Huber won, and is the loss every later round uses
4	31/32	matbench	Huber joint at three seeds; the first joint ensemble	recall@10% 70.1%, identical to the round-1 baseline
5	31/32	matbench	Scaled to eight seeds, forming three ensembles A/B/C	Ensembles 68.9–70.1%; established the seed spread
6	<i>No surviving artifacts — see the caveat box below. Recorded as a gap, not reconstructed.</i>			
7	31/32, 35_merge_aflow	matbench; matbench + 737 soft AFLOW	Third head, $\log_{10} \gamma$. Soft-material weighting $\kappa^{-\alpha}$ at $\alpha \in \{0, 0.4\}$, crossed with the two training sets	γ head works on AFLOW’s 1,022-crystal slice: γ MAE 0.36 $\rightarrow$ 0.19, r 0.33 $\rightarrow$ 0.57. Best recall 70.7% (+AFLOW, $\alpha=0.4$)
8	38_train_gamma_transfer	matbench $\rightarrow$ AFLOW	Two-stage transfer: freeze the matbench trunk, train only the γ head on AFLOW	Worse. Attributed at the time to a matbench/AFLOW “elastic-convention offset”; §13 shows that framing was wrong
9	37_train_gamma	AFLOW 5,563	Round 7’s recipe retrained single-stage on the full AFLOW set, three seeds	recall@10% 19.0 $\rightarrow$ 43.4/45.8/39.8%. But it underfits (train MAE on K 0.103), and its γ head collapsed to near-constant

Four things this table must not be read as saying

Round 6 is a genuine gap. No `history_*`, `summary_*` or prediction file carries an `r6` tag, and no script or document describes one. The project README refers to “rounds 2–6” collectively as a block of joint architecture, loss-weighting and augmentation variants, so round 6 was most likely one such variant that was run and not kept. It is left blank rather than invented.

The recall@10% column is on the superseded target. Every recall figure for rounds 1–7 is scored against this project’s own in-house κ_L reference, which was later found to be *circular* — it ranks predicted κ against a target built by the same Slack chain and the same derived γ , so γ cancels and a model with a genuinely better γ scores worse. Re-scoring against AFLOW-AGL’s independent κ_L lowered recall@10% by 20.5 points on average. The rows are comparable to *each other*, not to any AGL-scored number.

“Seven rounds never moved recall off 70.1%” is not quite true. That sentence appears in earlier drafts of this work. Fingerprinting the saved prediction vectors showed recall actually spans 64.0–70.7% across the 26 distinct runs, and round 7’s +AFLOW $\alpha=0.4$ arm reaches 70.7%, marginally *above* the baseline. The conclusion — that no round produced a gain worth having — stands; the constant 70.1% does not.

One number looked unresolved, and is now settled. `31_train_joint.py`’s docstring quoted the separate models’ residual correlation as +0.297 while the README’s table reported separate +0.263 against joint +0.297. Recomputing from the saved predictions (`60_resolve_residual_corr.py`) shows the docstring is right: +0.263 and +0.297 are *both* separate-model numbers, the single model and the 3-model ensemble. The README’s table therefore contained no joint model at all — it had taken the second and third columns of 31’s own correctly-labelled `this run | 1 model each | 3-model ens.` output and relabelled them “separate” and “joint”. The joint model’s real residual correlation is +0.457, 1.5 $\times$ the

13 What the two label sources actually disagree about

Round 8’s failure was attributed to an “elastic-convention offset” between matbench and AFLOW, and a single number — $|\Delta \log_{10} G| = 0.1085$ — was hard-coded into four scripts as *the* cross-source label noise. Re-measuring it from the raw label files (`59_cross_source_label_noise.py`) corrects three things.

It is not a convention difference. AFLOW’s own columns are `ael_bulk_modulus_vrh` and `ael_shear_modulus_vrh`: AEL is the *method*, a strain-based elastic library, and it reports VRH averages exactly as matbench does. Both numbers are the same physical quantity from two different DFT workflows, so the disagreement is workflow — k -points, strain magnitude, relaxation tolerance, functional — not definition.

0.1085 is the soft subset’s number, not the general one. On the full overlap the disagreement is roughly half as large, and it is centred scatter rather than a shift: subtracting the mean offset changes the MAE by 0.6%. The distribution is heavy-tailed — median $|\Delta|$ is 0.024 against an MAE of 0.054, with $p_{99} = 0.43$ — so most compounds agree to $\sim 5\%$ and a minority disagree by factors near three.

Table 10: Cross-source disagreement in G , by population and match rule. The soft-subset rows bracket the 0.1085 the project had been quoting; the full overlap does not come close to it.

Population	Match rule	n	$ \Delta \log_{10} G $	bias
AFLOW full (5,563)	formula	2,637	0.0612	−0.0032
AFLOW full (5,563)	formula+atoms	2,666	0.0539	−0.0056
AFLOW soft subset	formula	455	0.1122	−0.0440
AFLOW soft subset	formula+atoms	450	0.1044	−0.0466

The soft subset’s offset is a selection artifact. `afLOW_soft.csv` is selected because *AFLOW* reports a small modulus, which over-represents AFLOW’s own downward errors. Slicing the unselected full overlap by each source in turn makes this explicit: selecting on AFLOW $G < 20$ gives a bias of -0.027 , selecting on matbench $G < 20$ gives $+0.030$. The sign follows whichever source did the selecting, which is regression to the mean, not a property of either database.

The consequence for the “labels set the floor” claim

Step 49 divided the soft subset’s 0.1085 by the whole-test-set model error 0.0781 to get 1.39, and concluded that the labels rather than the network set the accuracy floor. The two halves describe different crystals, so that ratio was never like-for-like. Population-matched it is $0.0539/0.0781 = 0.69$ on the full set and $0.1044/0.1033 = 1.01$ on the soft set, and per stiffness band the model’s own error exceeds the label disagreement in *all* of them (ratios 0.68–0.86). On the moduli the network, not the labels, is the weaker half.

This does *not* rehabilitate AGL’s Grüneisen parameter. That comparison runs the other way — γ ’s disagreement is many times the error its head achieves — and correcting the moduli baseline downward makes γ look worse by comparison, not better. Note also that the γ figure rests on 43 Table 1 crystals, a third population again, so it remains indicative rather than like-for-like.

14 Limitations and next steps

- **Soft materials remain the weakest region.** Residuals fan out below ~ 30 GPa. This is partly genuine label noise — DFT is least reliable for soft, anharmonic crystals — but it is also precisely the ultralow- κ regime PINK targets, so it is the most valuable place to improve.

- **Only $\sim 11\text{k}$ labelled crystals exist.** Accuracy is now limited by data and label quality rather than by the model. Semi-supervised pre-training on unlabelled structures is the obvious lever.
- **κ_L error compounds stage 1’s.** Section 7’s 47% MAE against the reference pipeline is not a new failure mode — κ_L multiplies several quantities that each carry stage 1’s own error, so the same soft-material weakness that limits K and G limits κ_L more. Improving stage 1 remains the highest-leverage way to improve stage 2.
- **The uncertainty-calibration check (Section 7) is weak by construction.** Disagreement against the reference model entangles that model’s own idiosyncrasies with our error, and is not a clean measurement of either. Section 7.4’s real experimental data does not fix this: only 2 of the 45 Table 1 materials are genuinely unseen by our ensemble, far too few to check whether a $p05$ – $p95$ interval actually contains the true value at the right rate. A true calibration check would need real measured κ_L values for many more crystals our ensemble never trained on than currently exist in this project.
- **Table 1’s own κ_L numbers are not a fully-automated prediction.** Section 7.4 traces the gap against our model to the Grüneisen parameter specifically: the paper’s Table 1 uses AFLOW/experimental γ values, not ones derived from predicted elastic constants the way a real screening run (ours, and the paper’s own `pink_predict.py`) must. A learned or looked-up γ that scales to hypothetical GNoME candidates — rather than only to materials well-studied enough to have a literature value — is the single highest-leverage improvement identified in this entire reproduction, ahead of a better modulus model.

Reference. Liu, Y.; Wang, X.; Hao, Y.; Li, X.; Sun, J.; Lookman, T.; Ding, X.; Gao, Z. *PINK: physics-informed machine learning for lattice thermal conductivity*. J. Mater. Inf. 2025, 5, 12. [doi:10.20517/jmi.2024.86](https://doi.org/10.20517/jmi.2024.86)

Underlying data. Matbench elastic benchmarks `matbench_log_kvrrh` and `matbench_log_gvrh` (Dunn *et al.*, npj Comput. Mater. 2020), derived from the Materials Project.